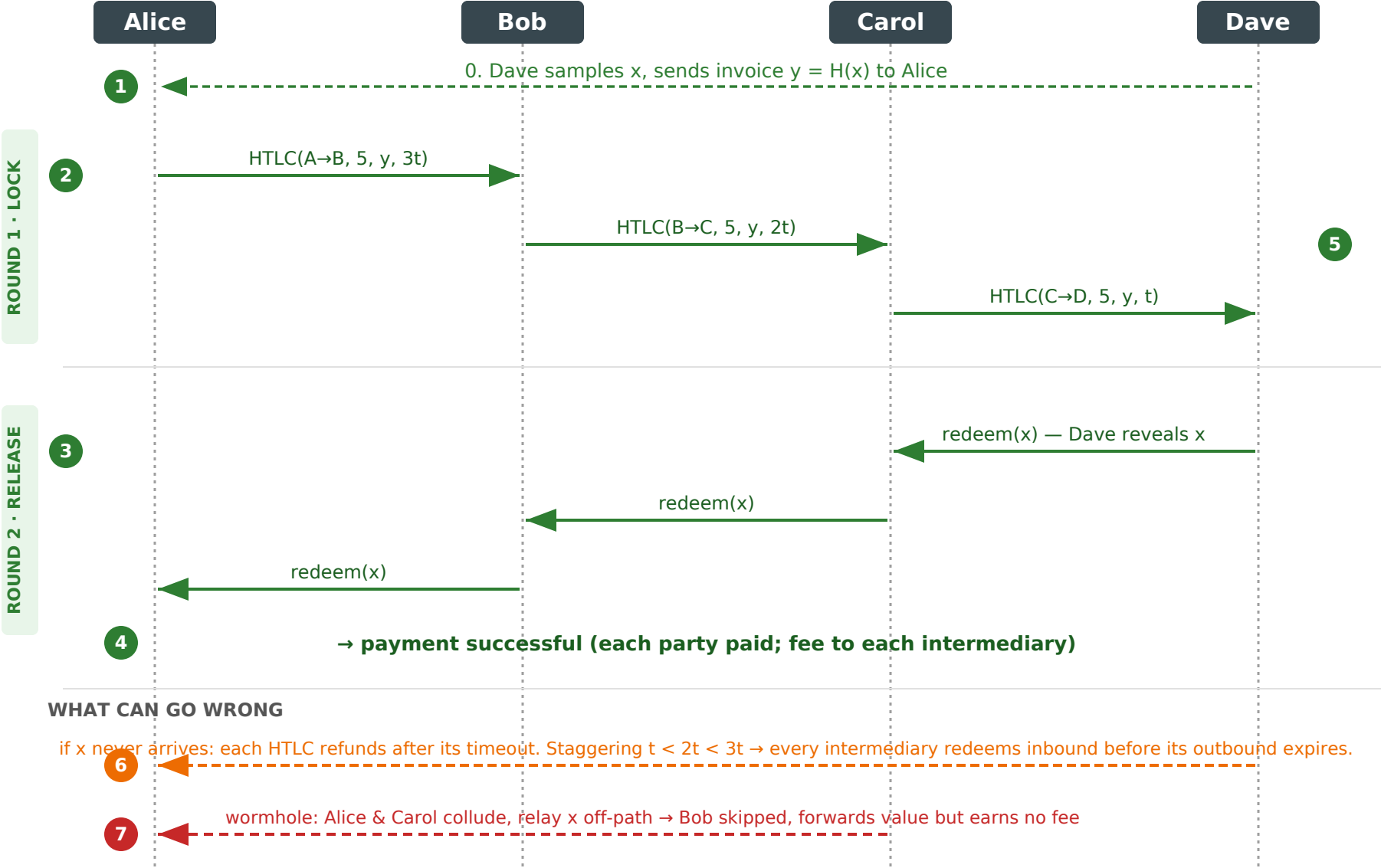


Multi-hop payment: Alice → Bob → Carol → Dave

annotated with multihop.spthy lemmas — see rail →



8 STAGGERED TIMELOCKS (block height →)



Rough draft — multihop.spthy lemmas mapped to the 4-party flow. Next: gaps / cltv\_blocks / t2b\_attack.

**LEMMAS · multihop.spthy**

- 1 **Invoice & preimage authentication**  
Invoice\_Authenticates\_Settlement · Preimage\_Secret\_Until\_Release · Forged\_Invoice\_Requires\_Key\_Compromise · Invoice\_Released ✓
- 2 **Lock causality (forward needs previous)**  
Forward1\_Requires\_Offer · Forward2\_Requires\_Forward1 · HTLC\_On\_Opened\_Channel ✓
- 3 **Release causality (redeem needs preimage)**  
Fulfill\_Requires\_Forward2 · Claim\_Requires\_Release · Settle\_Requires\_Receiver\_Release ✓
- 4 **Atomicity + non-vacuity**  
Payment\_Atomicity\_Under\_Liveness ✓  
Multihop\_Payment\_Possible ✓ (witness)
- 5 **Value & fee conservation**  
EndToEnd\_Value\_Conservation · Fee\_Conservation\_Hop1/2 · Fee\_Strictly\_Positive\_Hop1/2 · Receiver\_Paid\_Invoice\_Amount ✓
- 6 **Refund / griefing**  
Refund\_Requires\_Timeout · Loss\_Requires\_Inaction ✓  
Griefing\_Capital\_Lock\_Reachable ✓ (reachable DoS)
- 7 **Wormhole fee theft**  
Wormhole\_Fee\_Theft\_Reachable ✓  
Wormhole\_Steals\_Exactly\_The\_Fees ✓ (theft = path fees)
- 8 **Intermediary safety (under liveness T2b)**  
Intermediary\_Never\_Loses\_Under\_Liveness ✓  
T2b\_Counterexample\_Blocked ✓ (race blocked with T2b)

✓ = machine-checked in multihop.spthy (43/43).  
green = all-traces safety · orange = benign/reachable  
red = attack shown reachable (by design, no PTLC).

Channel-layer lemmas (open/update/close/punish) live on the Alice↔Bob figure.